

4.13.19

EE25BTECH11014 - Bhoomika Lokesh

Question: If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3:2$, then k equals

1) $\frac{29}{5}$

2) 5

3) 6

4) $\frac{11}{5}$

Solution:

Let the point **C** divide the line segment joining **A** and **B** in the ratio $3:2$ where

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1)$$

$$\mathbf{B} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (2)$$

By section formula,

$$\mathbf{C} = \frac{\lambda \mathbf{B} + \mathbf{A}}{(\lambda + 1)}. \quad (3)$$

$$\text{Here } \left(\lambda = \frac{3}{2} \right) \quad (4)$$

The equation of the line can be represented as:

$$\mathbf{n}^T \mathbf{x} = c_1 \quad (5)$$

$$\text{where } \mathbf{n}^T = \begin{pmatrix} 2 & 1 \end{pmatrix} \text{ and } c_1 = k. \quad (6)$$

As **C** lies on the above line,

$$\mathbf{n}^T \mathbf{C} = c_1 \quad (7)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \frac{\lambda \mathbf{B} + \mathbf{A}}{(\lambda + 1)} = k \quad (8)$$

$$\frac{1}{5} \begin{pmatrix} 2 & 1 \end{pmatrix} (3\mathbf{B} + 2\mathbf{A}) = k \quad (9)$$

$$\frac{1}{5} \begin{pmatrix} 2 & 1 \end{pmatrix} \left(\begin{pmatrix} 6 \\ 12 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right) = k \quad (10)$$

$$\frac{1}{5} \begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 14 \end{pmatrix} = k \quad (11)$$

$$k = \frac{16}{5} + \frac{14}{5} = 6. \quad (12)$$

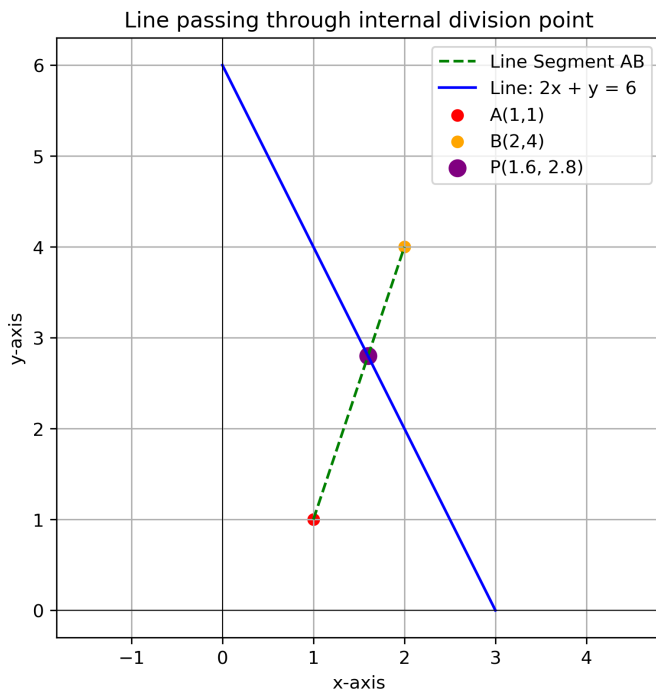


Fig. 4.1: Graph